Software Requirements Specification (SRS)

Project: Attrition Analysis System (AAS)

Version: 1.0

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Revision history

Version	Date	Author	Change summary	Approval
0.1	13-10-2026	Project team	Initial requirements from problem statement	-
1.0	27-10-2026	Project team	Complete SRS with use-case diagrams and RTM	Pending

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			
Project Guide			

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1. Introduction

1.1 Purpose

This document is the Software Requirements Specification (SRS) for the Attrition Analysis System (AAS), a web application that any organization can use to keep track of employees leaving the organization, understand why they leave and act early to retain people. It defines the functional and non-functional requirements, interfaces and verification criteria for developers, testers and evaluators.

1.2 Scope

AAS covers: maintenance of employee records (manual, CSV and HRMS sync); recording resignations and other separations with reasons; exit clearance tracking; exit-interview surveys; attrition analytics (rates, breakdowns, trends, reasons); flight-risk prediction for current employees; retention action tracking; threshold alerts and reports. The system is multi-tenant so several organizations can use one deployment with fully isolated data.

Out of scope: payroll and full-and-final settlement calculation, recruitment / backfilling, attendance and leave management, and the internal implementation of client HRMS products (only their APIs are used).

1.3 Audience

Developers, QA engineers, HR domain experts, system administrators, the project guide and assessment evaluators.

1.4 Definitions

Term	Meaning
Attrition	Employees leaving the organization for any reason in a period.
Attrition rate	(Separations in period / average headcount in period) x 100.
Separation	The event of an employee leaving (resignation, termination, retirement, contract end).
Voluntary / Involuntary	Employee-initiated exit / organization-initiated exit.
Regretted loss	A leaver the organization wanted to keep (e.g. high performer).
Flight risk	Predicted likelihood that a current employee will leave soon.
Tenant	One client organization using the system with isolated data.
Acronyms	AAS, HR, HRMS, CHRO, RBAC, MFA, TOTP, JWT, PII, CSV, API, TLS, RTO, RPO, WCAG, DPDP, RTM.

2. Overall description

2.1 Product perspective

AAS is a new, self-contained web application (SaaS). HR users access it through a browser; exiting employees access only a public, token-protected survey page. It integrates with an organization's existing HRMS for employee master data and with an email service for notifications. It does not replace the HRMS; it adds attrition-specific tracking and analytics on top of it.

2.2 Major product functions

- User authentication with MFA and role-based access
- Organization setup (departments, reason categories, alert threshold)
- Employee record management, CSV import and HRMS synchronisation
- Recording separations and tracking the exit clearance workflow
- Exit-interview surveys with reminders

- Attrition dashboard: rates, breakdowns, 12-month trends, exit-reason analysis
- Flight-risk scoring of current employees and retention action tracking
- Threshold alerts, PDF/CSV export and scheduled email reports

2.3 User roles and characteristics

Role	Characteristics / needs
HR Admin	Configures the organization, users and integrations; moderate technical skill.
HR Manager	Records separations, runs analyses, owns exit process; daily user, expects fast forms.
Department Manager	Views own team's attrition and risk, logs retention actions; occasional user.
Leadership / CHRO	Read-only dashboards and reports; wants summaries in a few clicks.
Exiting Employee	Fills the exit survey once via an emailed link; no account, no training.

2.4 Operating environment

Server: Linux containers (Docker) on a cloud VM or Kubernetes, Java 21 / Spring Boot, Python 3.12 for the risk model, PostgreSQL 16, Redis. Client: latest two versions of Chrome, Edge and Firefox on desktop or tablet.

2.5 Constraints

- Must comply with India's Digital Personal Data Protection (DPDP) Act, 2023 (and GDPR principles for international clients) since employee personal data is processed.
- Exit-survey feedback must remain anonymous at the manager level.
- HRMS integration limited to the REST APIs exposed by the client's HRMS.
- Project to be delivered within one academic semester using open-source technologies.

2.6 Assumptions and dependencies

- Organizations provide accurate employee data (join dates, departments, managers).
- An SMTP / email API account is available for notifications.
- Risk scoring needs at least 12 months of historical separations for meaningful results.

3. External interface requirements

3.1 User interfaces

- Responsive web portal with a left navigation bar: Dashboard, Employees, Separations, Surveys, Risk, Reports, Settings.
- Dashboard with KPI tiles (attrition %, voluntary %, regretted %, early attrition %), trend line chart, department bar chart and exit-reason chart, and global filters (period, department, location).
- Separation form on a single page with inline validation.
- Mobile-friendly public exit-survey page (no login).
- Accessibility: keyboard navigation, screen-reader labels, high-contrast colour palette (WCAG 2.1 AA).

3.2 Hardware interfaces

No special hardware. Users need a PC, laptop or tablet with a modern browser. The server runs on standard cloud virtual machines.

3.3 Software interfaces

- **Client HRMS REST API:** JSON over HTTPS for nightly employee sync (e.g. Zoho People, Keka, SAP SuccessFactors adapters).
- **Email service:** SMTP or transactional email API for survey links, alerts and reports.
- **PostgreSQL:** relational store for all tenant data; Redis for cached dashboard aggregates.
- **Risk Scoring Service:** internal REST API (Python, scikit-learn) for weekly scoring.

3.4 Communications

- HTTPS with TLS 1.2+ for all browser and service traffic; HSTS enabled.
- REST/JSON APIs versioned under /api/v1; JWT bearer tokens.
- Outgoing emails sent asynchronously via a queue with retry and exponential back-off.

4. System features (detailed)

Each requirement below has acceptance criteria and a reference test case. IDs follow AAS-F-###. In total the project defines 25 functional requirements, 7 non-functional requirements, 3 security objectives and 8 security requirements.

4.1 User Access & Organization Setup

Description: Securely sign users in, apply role-based access, and let each client organization configure itself (the system is multi-tenant so that any organization can use it).

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
AAS-F-001	The system shall authenticate users with email and password and shall require a 6-digit TOTP one-time code for HR Admin and HR Manager roles.	Functional	High	Security	AC-AAS-F-001: Valid credentials + valid OTP create a session; a wrong or expired OTP is rejected with a generic error. Test: TC-AUTH-01	Auth & RBAC Service
AAS-F-002	The system shall enforce role-based access (HR Admin, HR Manager, Department Manager, Leadership) and restrict a Department Manager to data of their own department.	Functional	High	Security / HR	AC-AAS-F-002: A Department Manager requesting another department's data receives HTTP 403; menus hide unauthorised features. Test: TC-AUTH-02	Depends on AAS-F-001
AAS-F-003	The system shall allow an HR Admin to register an organization and configure departments, locations, grades, exit-reason categories and the attrition alert threshold (%).	Functional	Medium	HR Admin	AC-AAS-F-003: Configured values appear in all dropdowns and the threshold is used by AAS-F-060. Test: TC-ORG-01	Org & User Mgmt

4.2 Employee Records

Description: Maintain the employee master data from which headcount and attrition are calculated.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
AAS-F-010	The system shall allow HR users to create, view, update and deactivate employee profiles (employee ID, name, department, designation, manager, join date, location, grade, performance rating, last promotion date).	Functional	High	HR	AC-AAS-F-010: Mandatory fields validated; duplicate employee ID within an organization rejected. Test: TC-EMP-01	Employee Service
AAS-F-011	The system shall import employee records in bulk from a CSV file of up	Functional	High	HR	AC-AAS-F-011: Valid rows imported; an error report	CSV template provided

	to 10,000 rows and report row-level validation errors without importing invalid rows.				lists row number and reason for each rejected row. Test: TC-EMP-02	
AAS-F-012	The system shall synchronise employee master data nightly from a connected HRMS through its REST API.	Functional	Low	IT	AC-AAS-F-012: New/changed HRMS records are reflected by the next morning; sync failures are logged and alerted. Test: TC-EMP-03	HRMS Connector; external API
AAS-F-013	The system shall store a monthly headcount snapshot per department to compute average headcount for any period.	Functional	High	Analytics	AC-AAS-F-013: Snapshot created on the 1st of each month; average headcount = mean of opening and closing snapshots. Test: TC-EMP-04	Scheduler

4.3 Separation (Exit) Management

Description: Record every employee who leaves, why they left, and the progress of their exit process.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
AAS-F-020	The system shall allow an HR Manager to record a separation with resignation date, last working day, separation type (voluntary / involuntary / retirement / end of contract), primary reason category and a regretted-loss flag.	Functional	High	HR	AC-AAS-F-020: Record saved, employee status changes to 'Notice period' and then 'Separated' after the last working day. Test: TC-SEP-01	Separation Service
AAS-F-021	The system shall reject a separation whose last working day is before the resignation date, or for an employee who is already separated.	Functional	High	HR	AC-AAS-F-021: Invalid input returns a field-level error and no record is created. Test: TC-SEP-02	Depends on AAS-F-020
AAS-F-022	The system shall generate an exit clearance checklist (asset return, knowledge transfer, access revocation, final settlement) and track each item until closure.	Functional	Medium	HR / IT	AC-AAS-F-022: Checklist created with every separation; the separation can be closed only when all items are complete. Test: TC-SEP-03	-
AAS-F-023	The system shall allow a pending resignation to be withdrawn and record it as a 'retained' case that is excluded from attrition counts.	Functional	Medium	HR	AC-AAS-F-023: Withdrawn case shows status 'Retained' and the attrition rate is unchanged. Test: TC-SEP-04	-

4.4 Exit Interview Survey

Description: Collect structured feedback from leavers to understand the real reasons for attrition.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
AAS-F-030	The system shall automatically email the exiting employee a one-time, tokenised exit-survey link when a separation is recorded.	Functional	High	HR	AC-AAS-F-030: Email dispatched within 5 minutes of AAS-F-020; link contains a random token (no personal data). Test: TC-SRV-01	Email Gateway
AAS-F-031	The system shall let the employee submit the survey (1-5 ratings for manager, pay, growth, work-life balance, culture, plus free text) without logging in, once per token, within 30 days.	Functional	High	HR	AC-AAS-F-031: Second submission or expired token is rejected; responses stored against the separation. Test: TC-SRV-02	-
AAS-F-032	The system shall send up to 2 reminder emails at 3-day intervals if the survey is not submitted.	Functional	Low	HR	AC-AAS-F-032: Exactly 2 reminders sent for a never-submitted survey; none after submission. Test: TC-SRV-03	Scheduler

4.5 Attrition Analytics & Dashboard

Description: Turn separation and headcount data into metrics and charts that explain who is leaving and why.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
AAS-F-040	The system shall compute the attrition rate for a selected period as (separations in period / average headcount in period) x 100 and show overall, voluntary and regretted attrition rates.	Functional	High	HR / Leadership	AC-AAS-F-040: For 12 separations and an average headcount of 240 the dashboard shows 5.00 %. Test: TC-ANA-01	Analytics Engine
AAS-F-041	The system shall show attrition broken down by department, location, tenure band (<1, 1-3, 3-5, >5 years), grade and manager, filterable by date range.	Functional	High	HR	AC-AAS-F-041: Each breakdown's separations sum to the overall total for the same filter. Test: TC-ANA-02	-
AAS-F-042	The system shall display a 12-month attrition trend chart compared with the same months of the previous year.	Functional	Medium	Leadership	AC-AAS-F-042: Chart shows 12 points per series; values match AAS-F-040 for each month. Test: TC-ANA-03	-
AAS-F-043	The system shall rank exit reasons by frequency and show average exit-survey scores per category.	Functional	High	HR	AC-AAS-F-043: Top reasons listed in descending order with counts and percentage share. Test: TC-ANA-04	Depends on AAS-F-031
AAS-F-044	The system shall report the early-attrition rate (employees leaving within 12 months of joining).	Functional	Medium	HR	AC-AAS-F-044: Rate computed only from separations with tenure < 12 months. Test: TC-ANA-05	-

4.6 Flight-Risk Prediction & Retention

Description: Predict which current employees are likely to leave so that managers can act before they resign.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
AAS-F-050	The system shall compute a weekly flight-risk score (0-100) for each active employee from tenure, months since last promotion, performance rating and the manager's team attrition, and classify it Low (<40), Medium (40-69) or High (>=70).	Functional	High	HR / Analytics	AC-AAS-F-050: Every active employee has a score dated within the last 7 days and a band matching the thresholds. Test: TC-RSK-01	Risk Scoring Service
AAS-F-051	The system shall display the top 3 factors contributing to each employee's risk score.	Functional	Medium	Department Manager	AC-AAS-F-051: Three factors with their relative contribution are shown on the employee risk card. Test: TC-RSK-02	-
AAS-F-052	The system shall allow managers to log retention actions (e.g. career discussion, compensation review) against an at-risk employee with owner, due date and outcome.	Functional	Medium	Department Manager	AC-AAS-F-052: Action saved, visible to HR, and overdue actions are highlighted. Test: TC-RSK-03	-

4.7 Alerts & Reports

Description: Notify the right people when attrition rises and share results outside the application.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
AAS-F-060	The system shall alert HR Managers and the relevant Department Manager (in-app and email) when a department's rolling 3-month attrition rate exceeds the configured	Functional	High	HR	AC-AAS-F-060: Alert generated once per breach, naming the department, rate and threshold. Test: TC-REP-01	Depends on AAS-F-003, AAS-F-040

	threshold.					
AAS-F-061	The system shall export any dashboard view as PDF and the underlying data as CSV.	Functional	Medium	Leadership	AC-AAS-F-061: Exported figures equal the on-screen figures for the same filters. Test: TC-REP-02	Report Service
AAS-F-062	The system shall email a scheduled monthly attrition summary report to subscribed users.	Functional	Low	Leadership	AC-AAS-F-062: Report delivered on the 2nd of each month to all subscribers. Test: TC-REP-03	Scheduler, Email Gateway

5. Non-functional requirements (detailed)

NFRs below are measurable and tied to the test plan. IDs AAS-NF-###.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
AAS-NF-001	The attrition dashboard shall load within 3 seconds (95th percentile) for an organization with 50,000 employee records and 5 years of history; other API calls shall respond within 500 ms (p95).	Performance	High	JMeter test on production-like data meets both targets. Test: TC-PERF-01
AAS-NF-002	The system shall support 200 concurrent users per organization and 100 organizations with less than 20 % increase in response time.	Scalability	High	Load test at target concurrency; p95 degradation < 20 %. Test: TC-PERF-02
AAS-NF-003	The system shall provide 99.5 % monthly availability (planned maintenance excluded) with RPO <= 24 hours and RTO <= 4 hours.	Reliability	High	Uptime monitoring report; restore drill from nightly backup completes within 4 h. Test: TC-OPS-01
AAS-NF-004	The user interface shall conform to WCAG 2.1 AA, and a new HR user shall be able to record a separation in under 3 minutes without training.	Usability / Accessibility	Medium	Accessibility audit (axe) passes; 4 of 5 test users complete the task in < 3 min. Test: TC-UX-01
AAS-NF-005	The service layer shall have at least 70 % unit-test coverage and each module shall be deployable without changes to other modules' code.	Maintainability	Medium	Coverage report from CI >= 70 %. Test: TC-QA-01
AAS-NF-006	The web application shall work on the latest two versions of Chrome, Edge and Firefox at 1280 px and tablet (768 px) widths.	Compatibility	Medium	Cross-browser checklist passes on all 6 browser versions. Test: TC-UX-02
AAS-NF-007	Personal data of separated employees shall be anonymised 7 years after the last working day, in line with the Digital Personal Data Protection (DPDP) Act, 2023.	Compliance / Data retention	High	Retention job anonymises eligible records in test data. Test: TC-SEC-06

5.1 Security

5.1.1 Security Objectives

- **SO-1 Confidentiality:** Employee personal data, exit-survey feedback and flight-risk scores are visible only to authorised users of the same organization; no organization can ever see another's data.
- **SO-2 Integrity & Accountability:** Separation records and the attrition figures derived from them cannot be altered without authorisation, and every change is traceable to a user and time.
- **SO-3 Anonymity of Feedback:** Leavers can give honest exit feedback without managers being able to identify individual responses.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
AAS-SR-001	TLS 1.2+ shall be mandatory for all network connections, with HSTS enabled.	Security	High	SSL scan shows no TLS < 1.2 and no weak ciphers. Test: TC-SEC-01
AAS-SR-002	Passwords shall be stored using bcrypt (cost >= 12); an account shall be locked for 15 minutes after 5 consecutive failed logins.	Security	High	DB contains only bcrypt hashes; 6th attempt is blocked. Test: TC-SEC-02

AAS-SR-003	Personal data fields (name, email, phone, salary band, gender) and survey free text shall be encrypted at rest using AES-256.	Security	High	Raw DB inspection shows ciphertext for these columns. Test: TC-SEC-03
AAS-SR-004	Every API request shall be authorised server-side by role and organization ID; a cross-tenant request shall return 404 without revealing the record's existence.	Security	High	Automated IDOR tests across two tenants all fail to access foreign data. Test: TC-SEC-04
AAS-SR-005	The system shall keep an append-only audit log of create/update/delete actions on employees, separations and user roles, retained for 7 years.	Security	High	Each action produces an audit entry; audit entries cannot be modified via the API. Test: TC-SEC-05
AAS-SR-006	Exit-survey responses shall be shown to Department Managers only as aggregates of at least 5 responses.	Security	High	A filter with fewer than 5 responses shows 'insufficient data'. Test: TC-SEC-07
AAS-SR-007	All inputs shall be validated and all queries parameterised (OWASP Top 10); CSV imports shall be sanitised against formula injection.	Security	High	OWASP ZAP scan has no High findings; '=cmd' cells are escaped. Test: TC-SEC-08
AAS-SR-008	Access tokens shall expire after 30 minutes of inactivity and logout shall revoke the refresh token.	Security	Medium	Expired / revoked tokens receive 401. Test: TC-SEC-09

6. Quality attributes & Acceptance tests

- Exit criteria for acceptance: all High-priority functional and security requirements implemented and verified, no Critical/High open defects, NFR-001 to NFR-003 met, and the RTM shows every test case passed.
- Acceptance test suites: Authentication & Access, Employee Records, Separation Management, Exit Survey, Analytics & Dashboard, Risk & Retention, Alerts & Reports, Performance, Security, Usability & Accessibility.
- Data accuracy check: attrition figures from AAS for a sample organization must match a manual spreadsheet calculation exactly (TC-ANA-01).

7. System models and diagrams

7.1 UML Use-Case diagrams

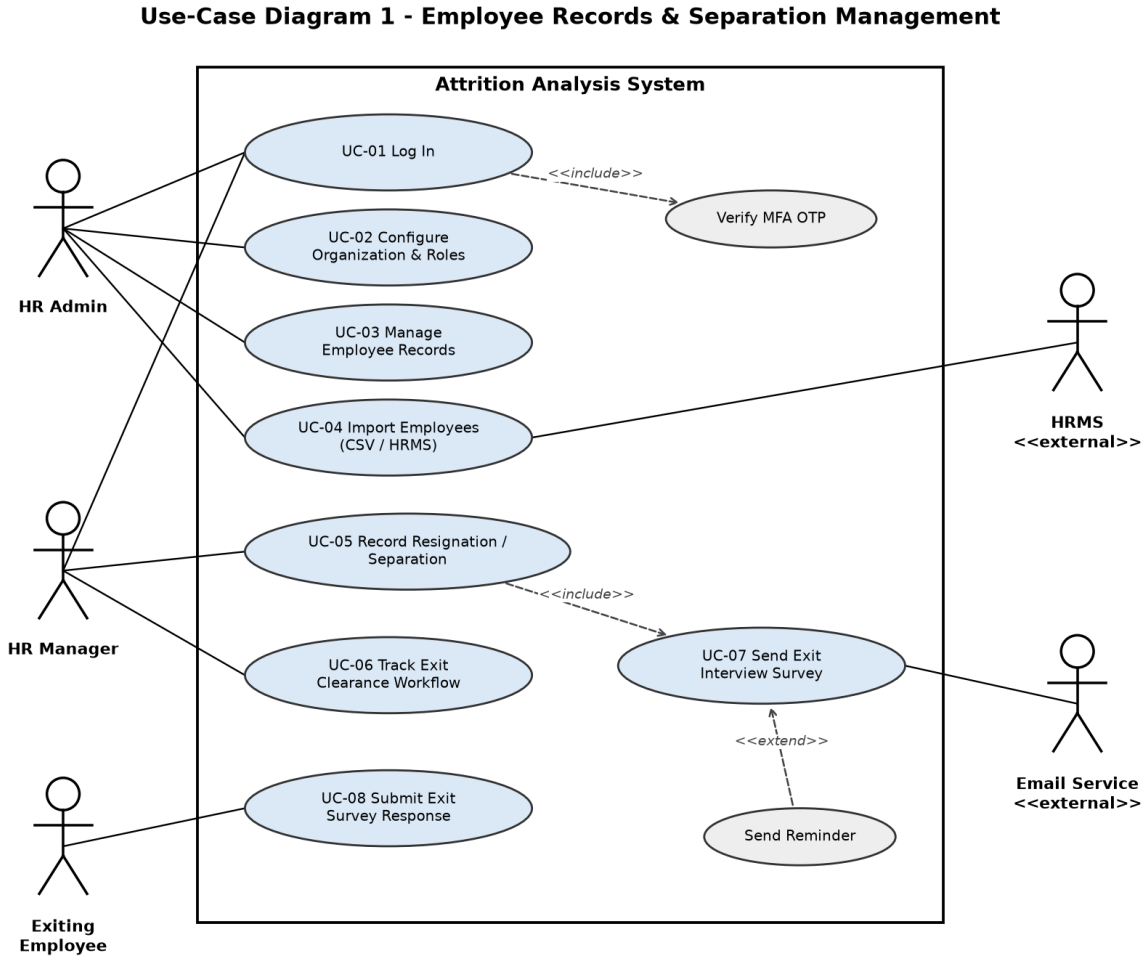


Figure 1 - Use-case diagram 1: employee records, separation and exit survey

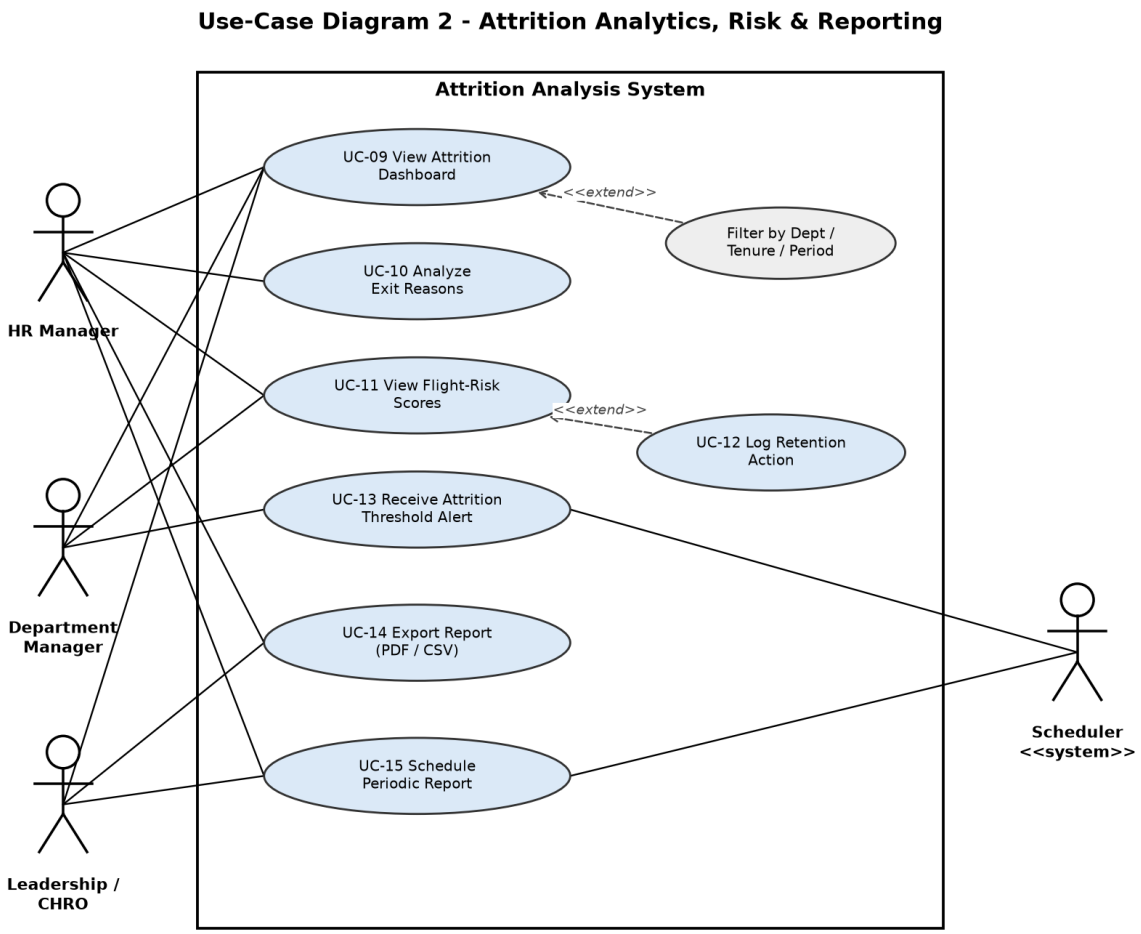


Figure 2 - Use-case diagram 2: attrition analytics, risk prediction, alerts and reporting

7.2 Use-case summary

Use case	Primary actor	Related requirements
UC-01 Log In (incl. Verify MFA OTP)	All HR users	AAS-F-001, F-002, SR-002, SR-008
UC-02 Configure Organization & Roles	HR Admin	AAS-F-003
UC-03 Manage Employee Records	HR Admin	AAS-F-010, F-013
UC-04 Import Employees (CSV / HRMS)	HR Admin, HRMS	AAS-F-011, F-012
UC-05 Record Resignation / Separation	HR Manager	AAS-F-020, F-021, F-023
UC-06 Track Exit Clearance Workflow	HR Manager	AAS-F-022
UC-07 Send Exit Interview Survey (ext. Send Reminder)	System, Email Service	AAS-F-030, F-032
UC-08 Submit Exit Survey Response	Exiting Employee	AAS-F-031, SR-006
UC-09 View Attrition Dashboard (ext. Filter)	HR Mgr, Dept Mgr, Leadership	AAS-F-040, F-041, F-042, F-044
UC-10 Analyze Exit Reasons	HR Manager	AAS-F-043
UC-11 View Flight-Risk Scores	HR Mgr, Dept Mgr	AAS-F-050, F-051
UC-12 Log Retention Action	Dept Manager	AAS-F-052
UC-13 Receive Attrition Threshold Alert	Scheduler -> Managers	AAS-F-060
UC-14 Export Report (PDF / CSV)	HR Mgr, Leadership	AAS-F-061
UC-15 Schedule Periodic Report	Leadership, Scheduler	AAS-F-062

8. Requirements Traceability Matrix (RTM)

Status: N = Not started, P = In progress, A = Accepted (tested and passed).

Req ID	Requirement short	Section ref / Design spec	Module	Test case(s)	Status (N/P/A)	Comments
AAS-F-001	authenticate users with email and password and...	4.1 / DS-Auth-01	Auth & RBAC Service	TC-AUTH-01	N	
AAS-F-002	enforce role-based access (HR Admin, HR Manager, Department...	4.1 / DS-Auth-01	Auth & RBAC Service	TC-AUTH-02	N	
AAS-F-003	allow an HR Admin to register an organization and configure...	4.1 / DS-Org-01	Org & User Mgmt	TC-ORG-01	N	
AAS-F-010	allow HR users to create, view, update and deactivate employ...	4.2 / DS-Emp-01	Employee Service	TC-EMP-01	N	
AAS-F-011	import employee records in bulk from a CSV file of up to 10...	4.2 / DS-Emp-01	Employee Service	TC-EMP-02	N	
AAS-F-012	synchronise employee master data nightly from a connected HR...	4.2 / DS-Emp-01	Employee Service	TC-EMP-03	N	
AAS-F-013	store a monthly headcount snapshot per department to compute...	4.2 / DS-Emp-01	Employee Service	TC-EMP-04	N	
AAS-F-020	allow an HR Manager to record a separation with resignation...	4.3 / DS-Sep-01	Separation Service	TC-SEP-01	N	
AAS-F-021	reject a separation whose last working day is before the res...	4.3 / DS-Sep-01	Separation Service	TC-SEP-02	N	
AAS-F-022	generate an exit clearance checklist (asset return, knowledg...	4.3 / DS-Sep-01	Separation Service	TC-SEP-03	N	
AAS-F-023	allow a pending resignation to be withdrawn and record it as...	4.3 / DS-Sep-01	Separation Service	TC-SEP-04	N	
AAS-F-030	automatically email the exiting employee a one-time, tokenis...	4.4 / DS-Srv-01	Survey Service	TC-SRV-01	N	
AAS-F-031	let the employee submit the survey (1-5 ratings for manager...	4.4 / DS-Srv-01	Survey Service	TC-SRV-02	N	
AAS-F-032	send up to 2 reminder emails at 3-day intervals if the surve...	4.4 / DS-Srv-01	Survey Service	TC-SRV-03	N	
AAS-F-040	compute the attrition rate for a selected period as (separat...	4.5 / DS-Ana-01	Analytics Engine	TC-ANA-01	N	
AAS-F-041	show attrition broken down by department, location, tenure b...	4.5 / DS-Ana-01	Analytics Engine	TC-ANA-02	N	
AAS-F-042	display a 12-month attrition trend chart compared with the s...	4.5 / DS-Ana-01	Analytics Engine	TC-ANA-03	N	
AAS-F-043	rank exit reasons by frequency and show average exit-survey...	4.5 / DS-Ana-01	Analytics Engine	TC-ANA-04	N	
AAS-F-044	report the early-attrition rate (employees leaving within 12...	4.5 / DS-Ana-01	Analytics Engine	TC-ANA-05	N	
AAS-F-050	compute a weekly flight-risk score (0-100) for each active e...	4.6 / DS-Risk-01	Risk Scoring Service	TC-RSK-01	N	
AAS-F-051	display the top 3 factors contributing to each employee's ri...	4.6 / DS-Risk-01	Risk Scoring Service	TC-RSK-02	N	
AAS-F-052	allow managers to log retention actions (e.g. career discuss...	4.6 / DS-Risk-01	Risk Scoring Service	TC-RSK-03	N	
AAS-F-060	alert HR Managers and the relevant Department Manager (in-ap...	4.7 / DS-Rep-01	Report & Notify Service	TC-REP-01	N	
AAS-F-061	export any dashboard view as PDF and the underlying data as...	4.7 / DS-Rep-01	Report & Notify Service	TC-REP-02	N	
AAS-F-062	email a scheduled monthly attrition summary report to subscr...	4.7 / DS-Rep-01	Report & Notify Service	TC-REP-03	N	
AAS-NF-001	Performance	5 / DS-Perf-01	Web UI / API / DB	TC-PERF-01	N	
AAS-NF-002	Scalability	5 / DS-Perf-01	Web UI / API / DB	TC-PERF-02	N	
AAS-NF-003	Reliability	5 / DS-Ops-01	Infrastructure	TC-OPS-01	N	
AAS-NF-004	Usability / Accessibility	5 / DS-UX-01	HR Web Portal	TC-UX-01	N	
AAS-NF-005	Maintainability	5 / DS-QA-01	All modules	TC-QA-01	N	
AAS-NF-006	Compatibility	5 / DS-UX-01	HR Web Portal	TC-UX-02	N	
AAS-NF-007	Compliance / Data retention	5 / DS-Sec-01	Security layer	TC-SEC-06	N	
AAS-SR-001	TLS 1.2+ shall be mandatory for all network connections...	5.1.2 / DS-Sec-01	Security layer	TC-SEC-01	N	
AAS-SR-002	Passwords shall be stored using bcrypt (cost >= 12); an...	5.1.2 / DS-Sec-01	Security layer	TC-SEC-02	N	
AAS-SR-003	Personal data fields (name, email, phone, salary band...	5.1.2 / DS-Sec-01	Security layer	TC-SEC-03	N	
AAS-SR-004	Every API request shall be authorised server-side by ro...	5.1.2 / DS-Sec-01	Security layer	TC-SEC-04	N	
AAS-SR-005	The system shall keep an append-only	5.1.2 / DS-Sec-01	Security layer	TC-SEC-05	N	

	audit log of creat...					
AAS-SR-006	Exit-survey responses shall be shown to Department Mana...	5.1.2 / DS-Sec-01	Security layer	TC-SEC-07	N	
AAS-SR-007	All inputs shall be validated and all queries parameter...	5.1.2 / DS-Sec-01	Security layer	TC-SEC-08	N	
AAS-SR-008	Access tokens shall expire after 30 minutes of inactivi...	5.1.2 / DS-Sec-01	Security layer	TC-SEC-09	N	